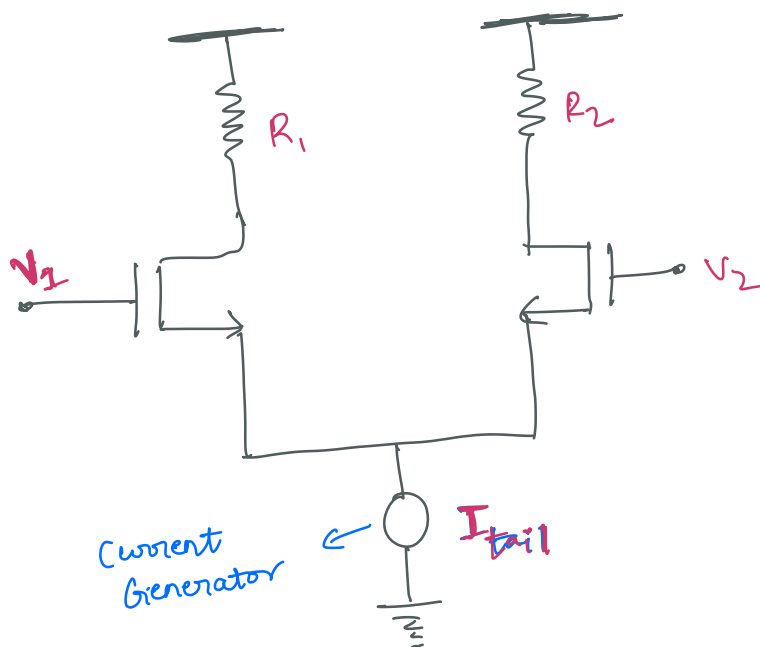


Lecture - 11
Differential Pair

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→ Basically Differential pair characterised by following Topology.



→ now if we suppose

$$V_1 = V_{10} + v_1$$

Nominal value

variation

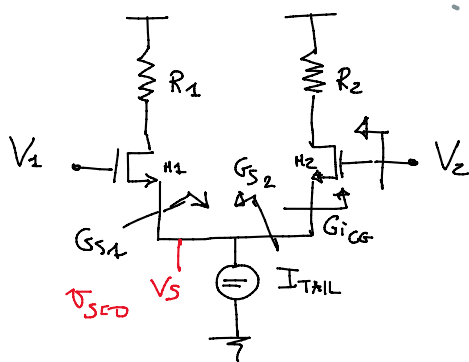
Similarly

$$V_2 = V_{20} + v_2$$

→ If we set a condition

$$V_{10} = V_{20} \quad \& \quad v_1 = -v_2 \triangleq \delta$$

i.e. $v_1 = \delta$
 $v_2 = -\delta$



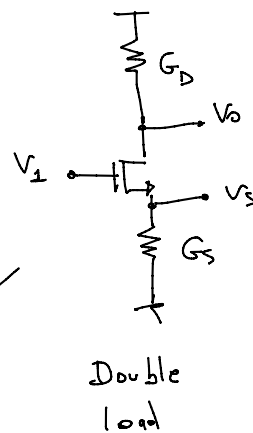
$$V_1 = V_{10} + v_1$$

$$V_2 = V_{20} + v_2$$

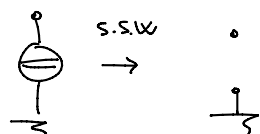
$$V_{10} = V_{20}$$

$$v_1 = -v_2 \triangleq \delta$$

$$\frac{v_s}{v_i} = \frac{g_m}{G_T + G_S + g_o \cdot \frac{G_S}{G_D}}$$



Double load



M1

$$G_S \equiv G_{icg}$$

$$G_{icg2} = \frac{G_{T2}}{1 + \frac{g_{o2}}{G_{L2}}}$$

$$\frac{v_s}{v_i} \bigg|_{v_2=0} \approx \frac{g_m}{G_T + G_S} = \frac{g_m}{G_T + G_{icg2}} = \frac{g_{m1}}{G_{T1} + \frac{G_{T2}}{1 + \frac{g_{o2}}{G_{L2}}}}$$

$$v_1 = -v_2 = \delta$$

$$\frac{v_s}{v_2} \bigg|_{v_1=0} = \frac{g_{m2}}{G_{T2} + \frac{G_{T1}}{1 + \frac{g_{o1}}{G_{L1}}}}$$

$$v_s = v_s \bigg|_{v_2=0} + v_s \bigg|_{v_1=0}$$

$$v_s = \left(\frac{g_{m1}}{G_{T1} + \frac{G_{T2}}{1 + \frac{g_{o2}}{G_{L2}}}} - \frac{g_{m2}}{G_{T2} + \frac{G_{T1}}{1 + \frac{g_{o1}}{G_{L1}}}} \right) \delta$$

$$v_s = 0$$

$$g_{o1} = \lambda_n \cdot I_D$$

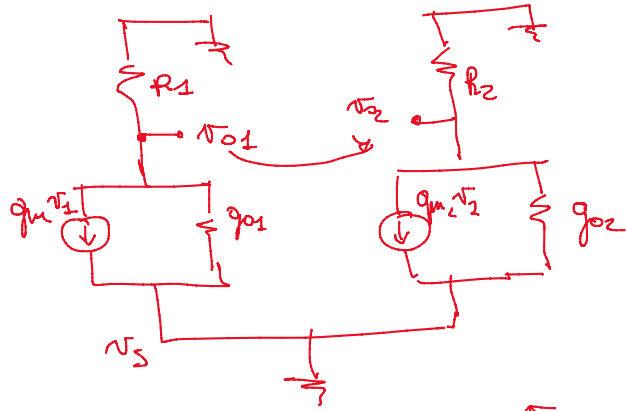
$$g_{m1,2} = \sqrt{2 \beta I_D}$$

$$\beta_1 = \beta_2$$

$$R_1 = R_2$$



SSC



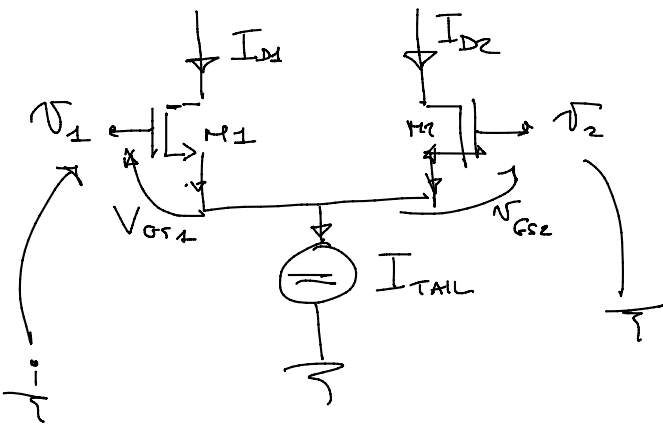
$$v_1 = \delta$$

$$v_2 = -\delta$$

$$v_{o1} = -\delta \cdot \frac{g_{m1}}{g_{o1} + G_1}$$

$$v_{o2} = -\frac{g_{m2}}{g_{o2} + G_2} (-\delta)$$

$$v_{o2} = v_{o1} = v_{od} = \left(\frac{g_{m2}}{g_{o2} + G_2} + \frac{g_{m1}}{g_{o1} + G_1} \right) \delta$$



$$\beta_1 = \beta_2 = \beta_n$$

$$v_{id} = v_1 - v_2 = v_{gs1} - v_{gs2} = (v_{gs1} - V_{TN}) - (v_{gs2} - V_{TN})$$

$$I_D = \frac{\beta}{2} (V_{GS} - V_T)^2 \Rightarrow V_{GS} - V_T = \sqrt{\frac{2 I_D}{\beta}}$$

$$v_{id} = \sqrt{\frac{2 I_{D1}}{\beta_1}} - \sqrt{\frac{2 I_{D2}}{\beta_2}} = \sqrt{\frac{2}{\beta_n}} (\sqrt{I_{D1}} - \sqrt{I_{D2}})$$

$$I_{D1} + I_{D2} = I_{TAIL}$$

$$I_{D1} = \frac{I_{TAIL}}{2} + \frac{I_{TAIL}}{2} \sqrt{\frac{\beta_n V_{id}^2}{I_{TAIL}} - \frac{\beta_n^2 V_{id}^4}{4 \cdot I_{TAIL}^2}}$$

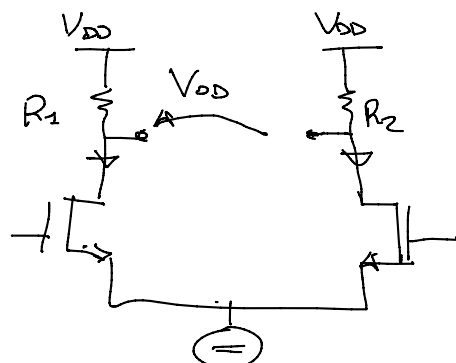
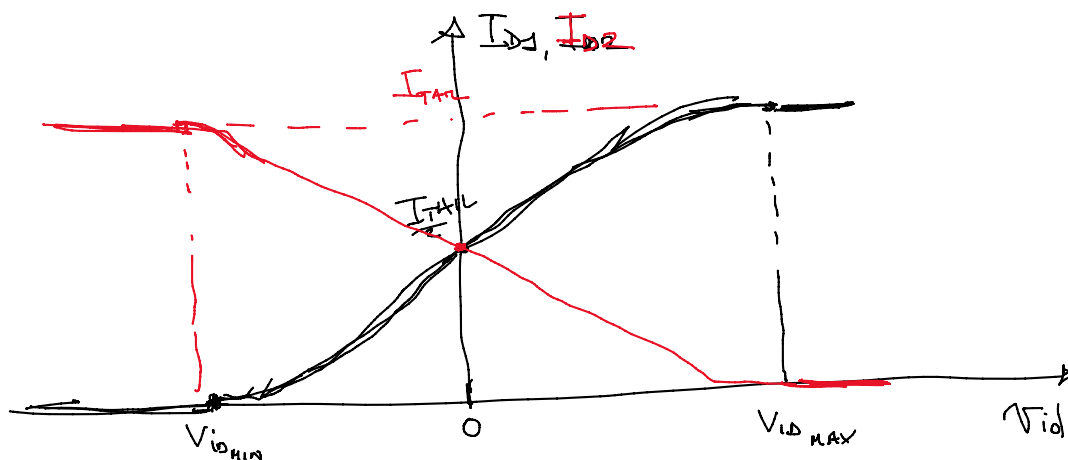
$$\rightarrow \begin{cases} I_{D1} = \frac{I_{TAIL}}{2} + \frac{I_{TAIL}}{2} \cdot \sqrt{\frac{1}{I_{TAIL}} - \frac{1}{4 \cdot I_{TAIL}^2}} \\ I_{D2} = \frac{I_{TAIL}}{2} - \frac{I_{TAIL}}{2} \cdot \sqrt{\frac{\beta_n V_{D1}^2}{I_{TAIL}} - \frac{\beta_n^2 V_{D1}^4}{4 \cdot I_{TAIL}^2}} \end{cases}$$

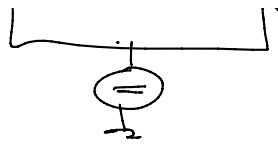
$$V_{ID} = 0 \Rightarrow I_1 = I_2 = \frac{I_{TAIL}}{2}$$

$$V_{ID} = ? \Rightarrow \begin{matrix} I_{D1} = I_{TAIL} \\ I_{D2} = 0 \end{matrix} \Rightarrow V_{ID_{MAX}} = \sqrt{\frac{2 I_{TAIL}}{\beta_n}} = \sqrt{2} \cdot \underbrace{\sqrt{\frac{2 I_{D10}}{\beta_n}}}_{V_{OV1}}$$

$$\begin{matrix} I_{D1} = 0 \\ I_{D2} = I_{TAIL} \end{matrix} \Rightarrow V_{ID_{MIN}} = -\sqrt{\frac{2 I_{TAIL}}{\beta_n}}$$

$$\boxed{\frac{I_{TAIL}}{2} = I_{D10}}$$

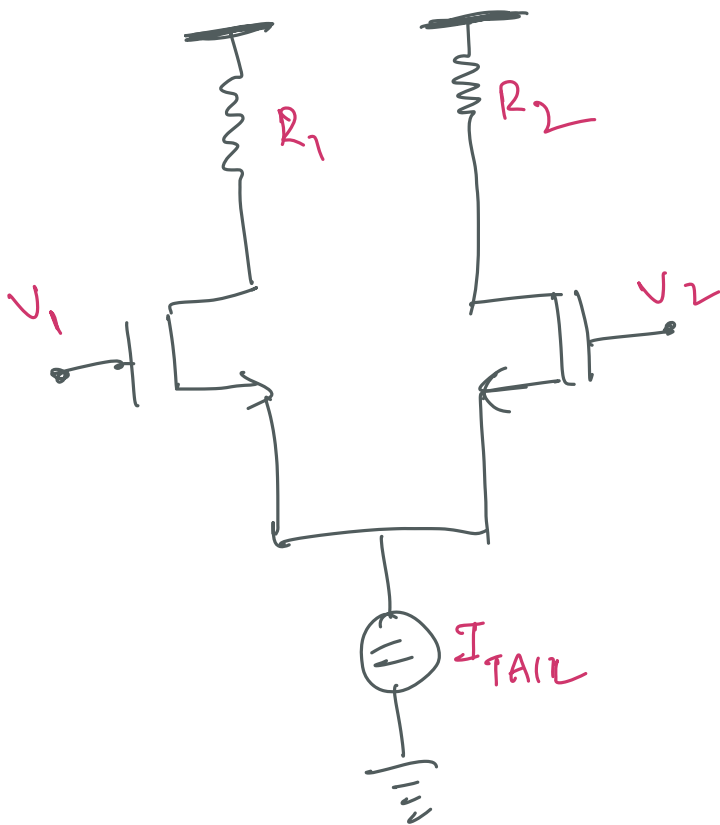




$$V_{oD} = V_{DD} - R_1 \cdot I_{D1} - (V_{DD} - R_2 \cdot I_{D2})$$

$$= -R_D \cdot (I_{D1} - I_{D2}) \quad \left(R_1 = R_2 \triangleq R_D \right)$$

Continuation of page 1 Notes.



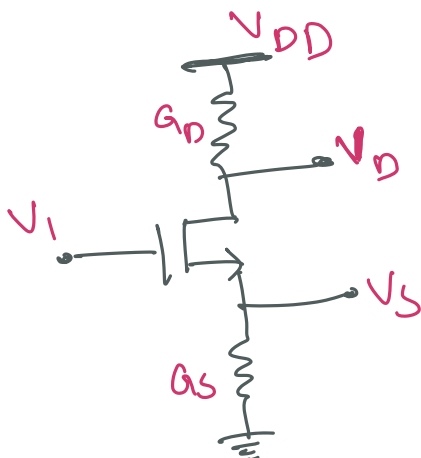
$$V_1 = V_{10} + v_1$$

$$V_2 = V_{20} + v_2$$

$$V_{10} = V_{20}$$

$$v_1 = -v_2 \triangleq v$$

→ In the previous class we studied about the Double load configuration if we compare it with differential pair circuit.

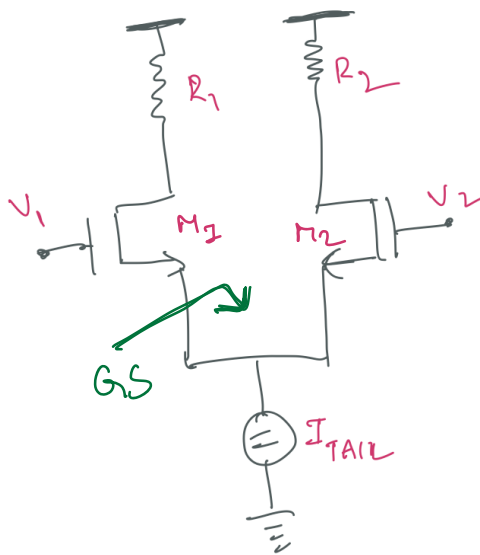


we got the ratios $\frac{V_S}{V_i}$, $\frac{V_O}{V_i}$

where
$$\frac{v_s}{v_1} = \frac{g_m}{G_T + G_S + g_o \cdot \frac{G_S}{G_D}}$$

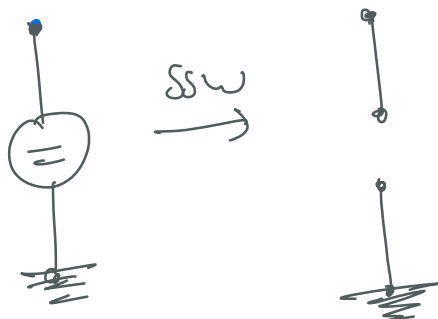
** → If we to imagine Differential pair as the Double load configuration where

$G = \frac{1}{R_1} = G_D$ & $G_S =$ The total conductance seen through the source of of the M_1 transistor.

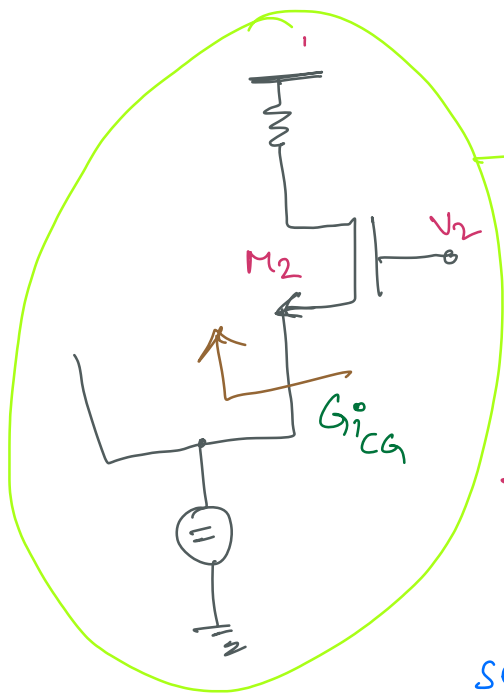


" G_S is the parallel b/w the conductance of "Ideal current" Generator and Transistor " M_2 " "

→ An Ideal current generator



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* This is the G_S seen through the source of M_1 which is parallel $4w$ Ideal current source & M_2 Transistor

* But if we look in the direction of the Brown arrow assuming $V_2 = 0$, what we see is the i/p conductance of a Common Gate.

→ Because, if we focus our analysis on M_1 we can apply super position principle and consider V_1 as the input and Make ZERO other i/p's. i.e. Make $V_2 = 0$, we are basically grounding the Gate of M_2 in ssw.

→ So G_S in this configuration w.r to M_1 is coincident with the i/p conductance of Common Gate.

$$G_S = \left(G_{icg} \parallel \frac{1}{\infty} \right)$$

↓

↓ conductance of

i/p conductance
or
 G_G

I deal current
source.

→ Using this $G_S = G_{ic} G_G$
we can substitution double load
equation for M_1 .

There is an other approximation

$$G_D \gg g_{DSAT} \quad (\text{Intrinsic o/p conductance of a device})$$

→

$$\frac{V_S}{V_i} \approx \frac{g_m}{G_T + G_S} = \frac{g_{m1}}{G_{T1} + G_{ic} G_2}$$

$\downarrow$
 $g_o \frac{G_S}{G_D}$

This term goes to zero as $G_D \gg g_{osat}$

→

$$G_{ic} G_2 = \frac{G_{T2}}{1 + \frac{g_{o2}}{G_{L2}}}$$

$$\Rightarrow \frac{V_S}{V_i} \approx \frac{g_{m1}}{G_{T1} + \frac{G_{T2}}{1 + \frac{g_{o2}}{G_{L2}}}}$$

$$\Rightarrow \left. V_s \right|_{\underline{V_2=0}} = V_1 \frac{g_{m1}}{G_{T1} + \frac{G_{T2}}{1+g_{o2}} \underline{G_{L2}}} \quad \text{--- ①}$$

$$V_s = \left. V_s \right|_{V_2=0} + \left. V_s \right|_{V_1=0}$$

Superposition principle

$$\left. V_s \right|_{V_1=0} = \frac{g_{m2}}{G_{T2} + \frac{G_{T1}}{1+\frac{g_{o1}}{G_{L1}}}}$$

This achieved by simply following the approach we had earlier.

for $\left. V_s \right|_{V_2=0}$

Input conductance of Transistor M_2 considering $V_1=0$

$$V_s = \left. V_s \right|_{V_2=0} + \left. V_s \right|_{V_1=0}$$

$$V_o = g_{m1} \cdot V_1 + g_{m2} \cdot V_2$$

$$\Rightarrow \frac{G_{T1} + \frac{G_{T2}}{1 + \frac{g_{o2}}{G_{L2}}}}{G_{T2} + \frac{G_{T1}}{1 + \frac{g_{o1}}{G_{L1}}}}$$

But $V_1 = -V_2 = \delta$

$$V_s = \left(\frac{g_{m1}}{G_{T1} + \frac{G_{T2}}{1 + \frac{g_{o2}}{G_{L2}}}} - \frac{g_{m2}}{G_{T2} + \frac{G_{T1}}{1 + \frac{g_{o1}}{G_{L1}}}} \right) \delta$$

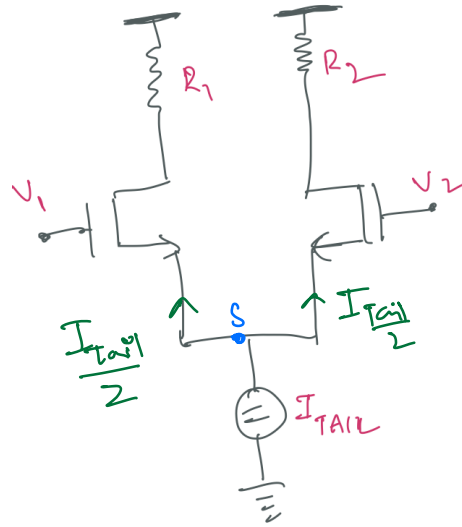
→ we know $g_m = \sqrt{2\beta I_D}$

If Transistors are equally sized then

$$\beta_1 = \beta_2$$

& if $R_1 = R_2$

→ what we can conclude is the fact that the current I_{tail} is splitted into the equal branches equally.



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*** and we can also see for $R_1 = R_2, R_1 = R_2$
 v_s will become zero.

$$\therefore v_s = \left[\frac{g_{m2}}{G_{T1} + \frac{G_{T2}}{1 + \frac{g_{o2}}{G_{L2}}}} - \frac{g_{m2}}{G_{T2} + \frac{G_{T1}}{1 + \frac{g_{o1}}{G_{L1}}}} \right] \delta$$

i.e. $v_s = 0 \rightarrow$ little $v_s = 0$

$\rightarrow$ So we have sort of Virtual Ground at

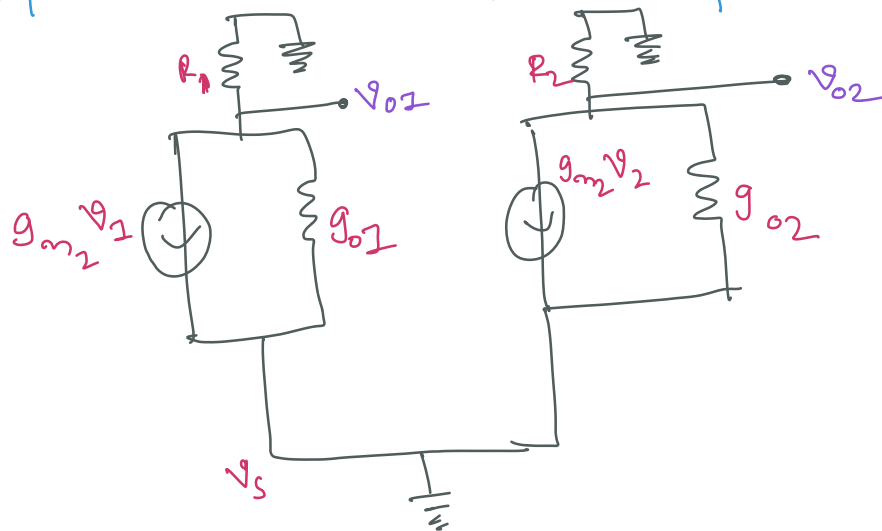
"S" $v_s = \text{Variation of voltage at "S"} = 0$

even if v_s is not equal to zero

it remains constant because $v_s = 0$

→ different ←

If we have a Virtual Ground at 'S'



$$v_1 = \delta, \quad v_2 = -\delta$$

→ Then

$$v_{o1} = -\delta \frac{g_{m1}}{g_{o1} + G_1} \quad \because v_1 = \delta$$

$$v_{o2} = \frac{-g_{m2}(-\delta)}{g_{o2} + G_2} \quad \because v_2 = -\delta$$

$$v_s = 0 \quad \text{when} \quad \begin{aligned} v_1 &= \delta \\ v_2 &= -\delta \end{aligned}$$

~~***~~ If we compute the voltage Drop
b/w V_{o1} & V_{o2}

$$\Rightarrow V_{od} = V_{o2} - V_{o1}$$

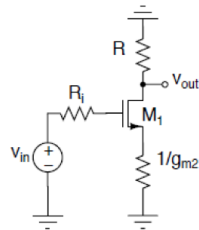
$$V_{od} \Rightarrow \left(\frac{g_{m2}}{g_{o2} + G_2} + \frac{g_{m1}}{g_{o1} + G_1} \right) f$$

This is basically the link V_{od} voltage
applied to our inputs and Differential quantity.

This is the SSA of the circuit.

→ Let's study how & why the devices
are useful (Diff-pair)

Toward a New Circuit



- If R_s is small enough (e.g. $1/g_{m2}$) the global G_m halves the one of the CS

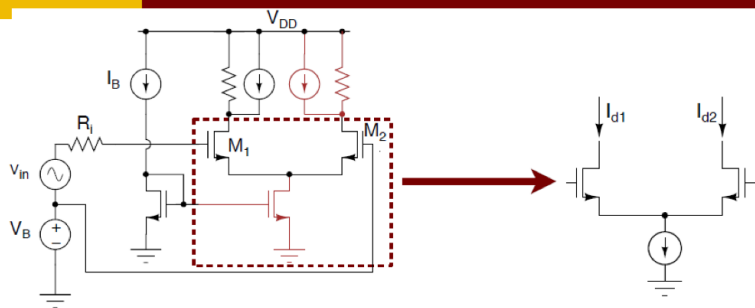
$$G_m = \frac{g_{m1}}{1 + g_{m1} \times \frac{1}{g_{m2}}} = \frac{g_m}{2}$$

$$\Rightarrow a_{v0} = G_m \times R = -\frac{1}{2} g_m R$$

- It looks like we are losing half of the gain.
- Is there any way we can retrieve that other half?

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Birth of Differential Pair

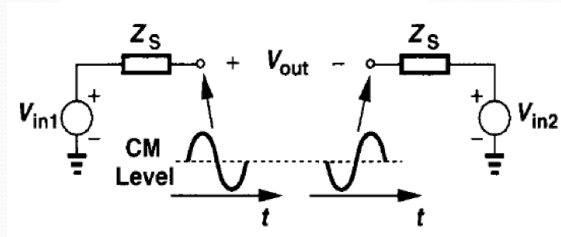
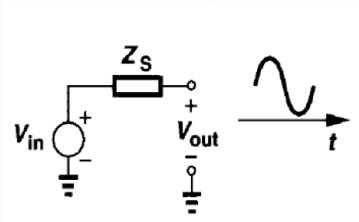


- The structure on the right is called the “differential pair,”
 - Its main feature is to evaluate the difference between two voltages.
 - In our circuit, these two voltages are V_B and $V_B + V_{in}$
 - To first order, changes in V_I and V_t do not affect the output voltage
- Ubiquitous in integrated circuits (not so popular in discrete implementation).

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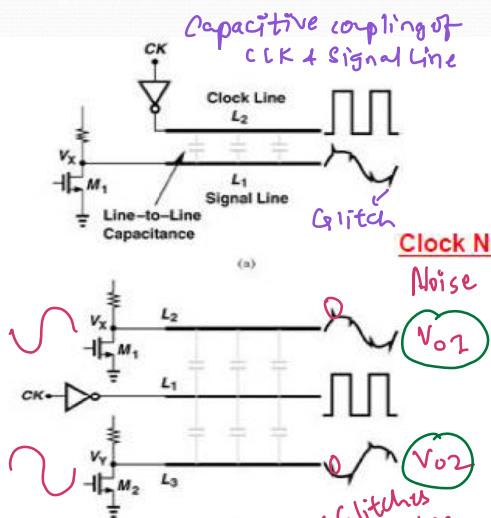
* It is much better to process Differential Signals than Single signal.

Single Ended and Differential Signals

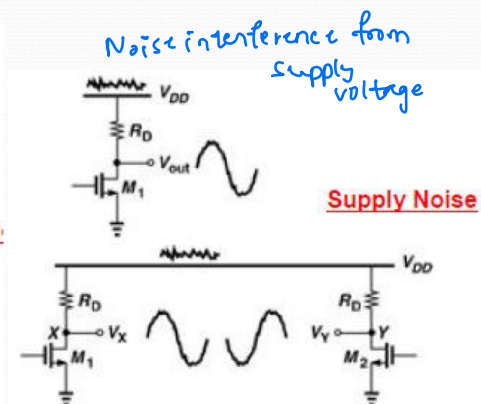


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Differential vs Single-Ended



Same Glitches induced by Clock line in both signals



RZ

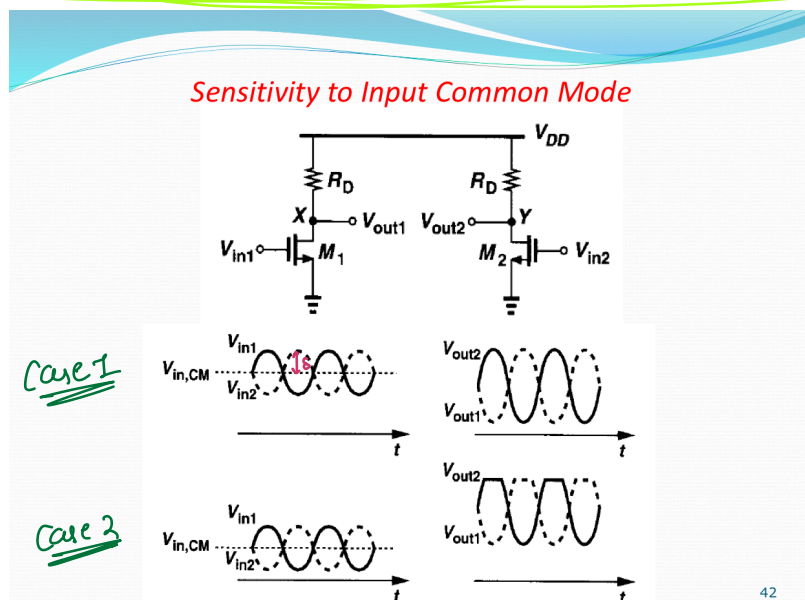
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→ So the Glitches in the both signals because of CLK are in the same direction.

→ So if we were to subtract the two signals V_{o1} & V_{o2} we would have two advantages. The subtraction doubles the information. While due to same sign of Glitches the Glitches are deleted.

→ So, basically we have Amplification of signal & Cancellation of Corruption (Glitches) simultaneously.

* $\therefore$ It is very useful to do this kind of *
* Differential processing on Signals. *



The Goal is to

Maximise

$$V_{o2} - V_{o1}$$

&

Minimize

$$\frac{V_{o2} + V_{o1}}{2}$$

$= V_{in \text{ common mode}}$

Case 1

$V_{in \text{ common mode}}$

$$= \frac{V_{o2} + V_{o1}}{2}$$

average of V_{in1} & V_{in2}

In this case sinusoid is correctly mirrored in the o/p.

$\therefore$ what we have is $V_{o2} - V_{o1}$ is the amplified version of the differential i/p.

$\rightarrow$ But the situation is radically different in case 2.

Case-2

the value of

V_{inCM}

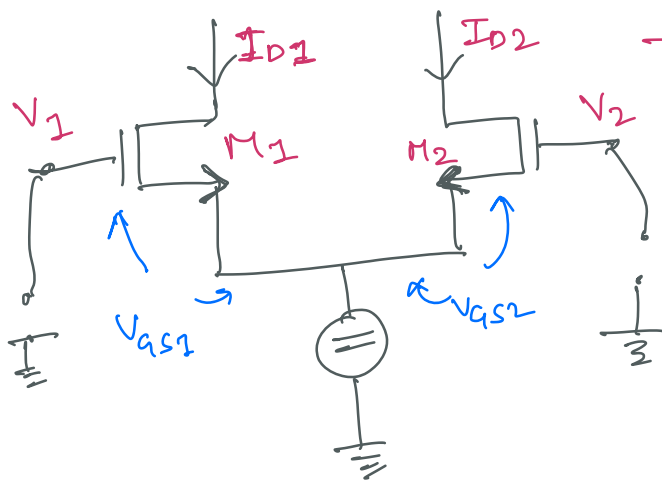
might be lower.

which may be responsible for clipping of the o/p.

$\therefore$ It is important to study Biasing of the Differential Amplifier because it may

effect Global Result.

* "To perform this Common Mode Analysis and to investigate the working regions of our device. let's do calculations."



→ for now neglect the kind of load connected to the transistors

$V_{GS} \rightarrow$ large signal
 $v_{gs} \rightarrow$ small signal

→ V_{GS1} & V_{GS2} are simply related to $V_1 + V_2$

$$V_{id} = V_1 - V_2 = V_{GS1} - V_{GS2} \quad \text{--- (a)}$$

Assuming the transistors are in saturation

$$I_D = \frac{\beta}{2} (V_{GS} - V_T)^2$$

$$\Rightarrow V_{GS} - V_T = \sqrt{\frac{2I_D}{\beta}}$$

Improrised version of (a)

$$V_{id} = (V_{GS1} - V_{TN}) - (V_{GS2} - V_{TN})$$

$$= \sqrt{\frac{2 I_{D1}}{\beta_1}} - \sqrt{\frac{2 I_{D2}}{\beta_1}}$$

for M_1 & M_2 in Saturation

$$V_{id} = \sqrt{\frac{2 I_{D1}}{\beta_n}} - \sqrt{\frac{2 I_{D2}}{\beta_n}}$$

if Transistors are identical

$$\beta_1 = \beta_2$$

$$V_{id} \Rightarrow \sqrt{\frac{2}{\beta}} (\sqrt{I_{D1}} - \sqrt{I_{D2}})$$

we have

$$I_{D1} + I_{D2} = I_{Tail} \quad \text{from the circuit}$$

~~*~~
~~*~~

$$I_{D1} = \frac{I_{Tail}}{2} + \frac{I_{Tail}}{2} \sqrt{\frac{\beta_n V_{id}^2}{I_{Tail}} - \frac{\beta_n^2 V_{id}^4}{4 \cdot I_{Tail}^2}}$$

$$I_{D2} = \frac{I_{Tail}}{2} - \frac{I_{Tail}}{2} \sqrt{\frac{\beta_n V_{id}^2}{I_{Tail}} - \frac{\beta_n^2 V_{id}^4}{4 \cdot I_{Tail}^2}}$$

~~*~~
~~*~~

This says that when $V_{id} = 0$

then

$$I_{D1} = I_{D2} = \frac{I_{Tail}}{2}$$

→ Then it would be also interesting to consider the opposite case, when at a certain point when we apply differential input i.e. $V_1 \neq V_2$ which implies currents $(I_{D1} + I_{D2})$ will also be different and at a certain point all the current I_{Tail} will flow in a single branch.

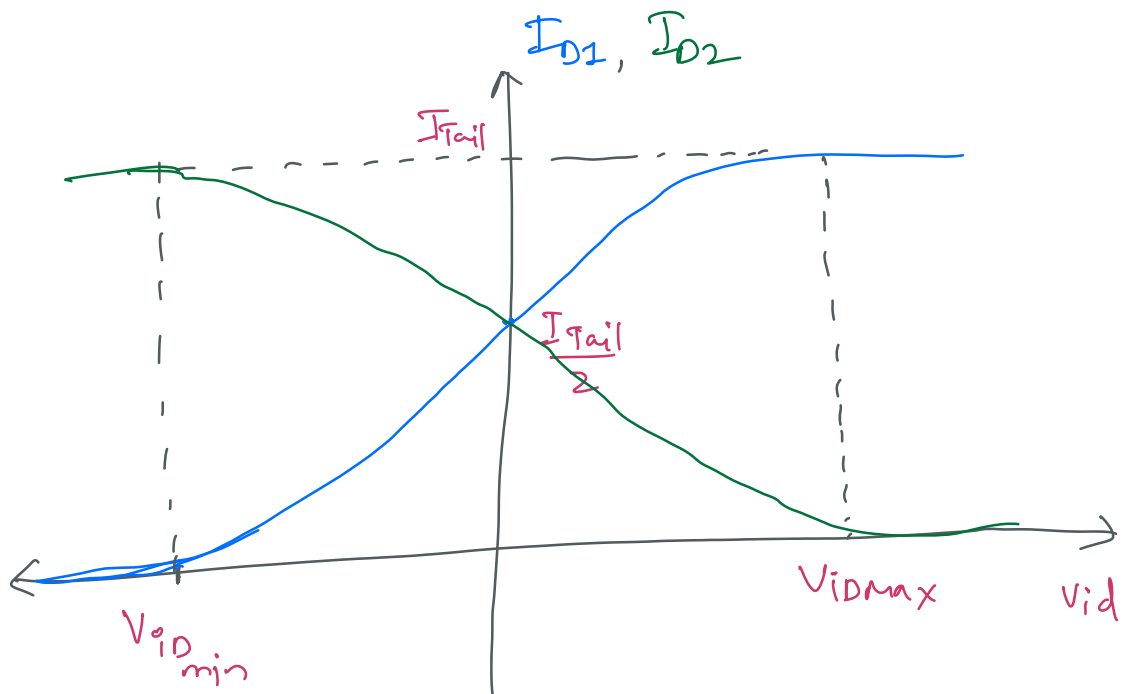
Q What is the Differential input necessary to create this full unbalance?
(i.e. I_{Tail} will flow through only one branch)

Ans we have

(i) $I_{D1} = I_{Tail} + I_{D2} = 0$ Simultaneously
when $V_{iD} = \sqrt{\frac{2 I_{Tail}}{\beta_n}}$

(ii) $I_{D1} = 0, I_{D2} = I_{Tail}$

$$V_{iD} = -\sqrt{\frac{2 I_{Tail}}{\beta_n}}$$



→ $\frac{I_{Tail}}{2} = I_{D10}$ i.e. when $V_{id} = 0$

using this we can write

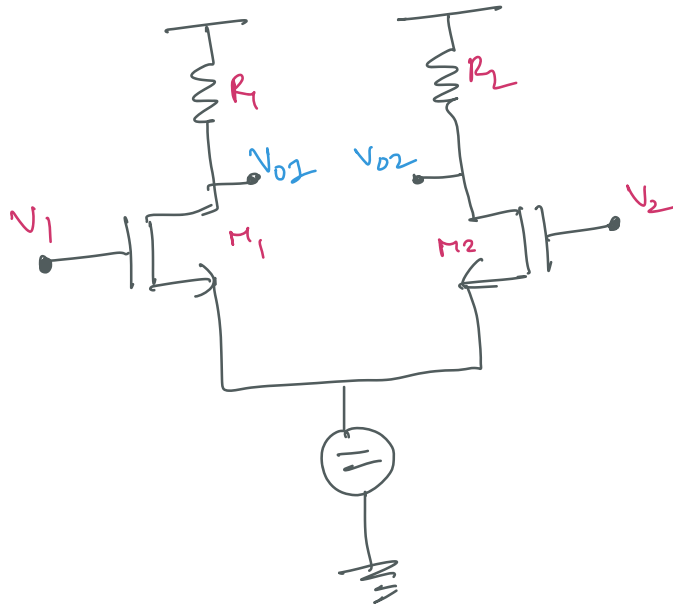
$$V_{id_{max}} = \sqrt{\frac{2I_{Tail}}{\beta_n}} = \sqrt{2} \sqrt{\frac{2I_{D10}}{\beta_n}}$$

Overdrive of Transistor 1

$$V_{id_{max}} = \sqrt{2} \cdot V_{ov1}$$

→ Till now we neglected the effect of

Load.



But, Basically Load plays no role in the Analysis we went through.

→ In Differential pair the importance of Load lies in the translation of the informative content related to currents into something related to voltages at V_{o1} & V_{o2}

$$\text{i.e. } \begin{cases} V_{o1} = V_{DD} - R_1 I_{D1} \\ V_{o2} = V_{DD} - R_2 I_{D2} \end{cases}$$

$$V_{DD} = V_{o1} - V_{o2}$$

$$= V_{DD} - R_1 I_{D1} - (V_{DD} - R_2 I_{D2})$$

$$\Rightarrow V_{OD} - R_1 I_{D1} - V_{OD} + R_2 I_{D2}$$

$$V_{OD} \Rightarrow R_2 I_{D2} - R_1 I_{D1}$$

if $R_1 = R_2 = R_D$

$$V_{OD} = R_D (I_{D2} - I_{D1})$$

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